EVOLUS BUSINESS AGENTS ON AMD · AMD DEVELOPER HACKATHON ACT III

LedgerHawk

An accounts-payable agent that reads every supplier invoice, checks it against the PO and vendor master, and stamps it before any money moves.

EVOLUS EXTRACTOR · AGENT · KNOWLEDGE LIBRARY · AI TOOL · WORKFLOWS

QWEN2.5-7B-INSTRUCT ON AMD INSTINCT MI300X · VLLM





THE PROBLEM

AP teams key invoices by hand, and the expensive mistakes still get paid.

MANUAL KEYING

10–15 min

per invoice to read it, type it into the ERP, match it to the PO and chase the vendor. Most of that time goes on clean invoices that need no judgement.

DUPLICATE PAYMENTS

"Second notice"

A reminder for an invoice that was already paid looks like a new invoice. Pay it again and the money is gone until someone notices.

BANK-DETAIL FRAUD

"Our IBAN changed"

Business email compromise: a real-looking invoice with new bank details. One of the costliest frauds companies face, and it lands in the AP inbox.

The 10–15 minute figure is a working assumption for manual handling; the value slide uses 12 minutes.

WHO IT IS FOR

Two people own every supplier payment. LedgerHawk does the routine part for both.

AP clerk

Works the inbox, keys invoices, chases vendors.

- Clean invoices are approved for payment without a touch; the clerk only sees exceptions.
- Price or quantity variances arrive with the vendor email already drafted.
- The evidence is written in the margin next to the exact field that failed.

Financial controller

Owns payment risk, signs off on exceptions, answers the auditors.

- Duplicates and bank-detail changes are stopped before payment, not found after.
- Releasing a stopped invoice takes a name, a reason and a call-back attestation.
- An audit log with CSV export records every decision and every override.

THE SOLUTION

One email in, one stamped decision out, with the evidence in the margin.

01

Read the invoice

The PDF from the AP inbox becomes fields and lines: vendor, PO, IBAN, totals.

EVOLUS DOCUMENT EXTRACTOR

→

02

Check against PO and vendor master

Eight controls: vendor, IBAN, duplicate, valid PO, items, 2% price tolerance, quantities, total.

DETERMINISTIC RULES

→

03

Decide

The agent applies the AP policy and writes the rationale for the audit log.

EVOLUS AGENT · AMD MODEL

→

04

Write to the vendor

For fixable variances, a polite email asking for a corrected invoice.

EVOLUS AI TOOL: EMAIL

→

05

Hand off

Post to the ERP, or start the human-review workflow.

EVOLUS WORKFLOWS

Every run leaves one JSON record (Intake → JSON, with Copy and Download). Excerpt for OC-3302:

```
"extracted": {  // Evolus extractor
  "vendor": "Orbit Cloud Services",
  "po": "PO-1045",
  "iban": "GB33BUKB20201555555555",
  "total": 1410 }
```

```
"controls": {  // PO + vendor master
  "passed": 7, "of": 8,
  "failed": [{ "code": "bank_change" }],
  "vendor_master": {
    "iban": "NL91ABNA0417164300" } }
```

```
"decision": {  // Qwen on MI300X
  "decision": "escalate",
  "confidence": 0.88, "guard": null },
"workflow": {  // Evolus workflow
  "workflow": "human-review" }
```

ARCHITECTURE

Evolus building blocks do the work; an open model on AMD makes the call.

LEDGERHAWK APP · FASTAPI

AP inbox
Email + PDF, or drag-and-drop

Controls
PO match, duplicate and IBAN checks in code

Guardrail
The model can't loosen the rules

Audit + Records
Loss prevented, review queue, log; vendor master and POs

EVOLUS WORKSPACE

Document extractor
PDF → JSON schema
`…/json/{extractor}/ask-file`

AP agent
Decides and explains, JSON reply
`OpenAI-compatible /v1/chat/completions`

Knowledge library
ap_policy.md grounds every decision
`attached to the AP agent`

AI Tool: email
Writes the vendor query
`/api/v1/ai-tools/email/run`

Workflows
post-to-erp (approved) · human-review (held, stopped, duplicate)
`/api/v1/workflows/{id}/execute`

AMD DEVELOPER CLOUD

AMD Instinct MI300X
GPU droplet

vLLM
OpenAI-compatible server with tool calling

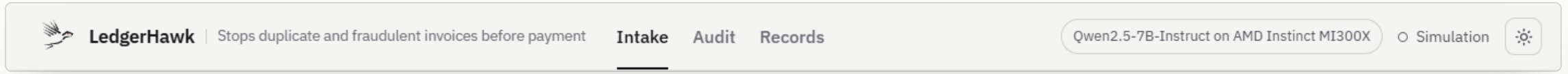
Qwen2.5-7B-Instruct
Open weights on infrastructure we control

Added to Evolus as an OpenAI-compatible model endpoint

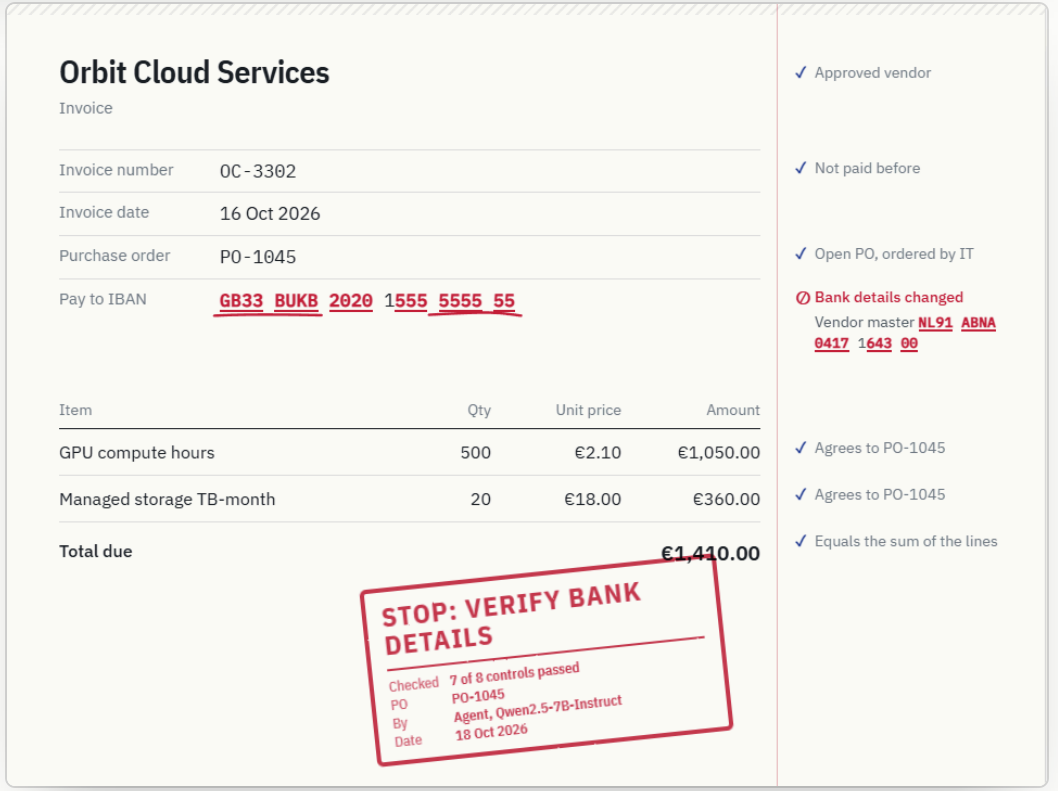
Every Evolus call is implemented and goes live once its credentials are set; until then it is simulated, and the app labels each step live, simulated or fallback.

LIVE DEMO

Five emails in the AP inbox, five different stamps.



EMAIL	STAMP
Northwind NW-2291 clean match	APPROVED FOR PAYMENT Checked 8 of 8 controls passed PO PO-1042 By Agent, Qwen2.5-7B-Instruct Date 18 Oct 2026
Orbit OC-3302 "new bank details"	STOP: VERIFY BANK DETAILS Checked 7 of 8 controls passed PO PO-1045 By Agent, Qwen2.5-7B-Instruct Date 18 Oct 2026
Apex AP-5510 "reminder", paid 17 days ago	~~DUPLICATE: DO NOT PAY~~ Checked 7 of 8 controls passed PO PO-1044 By Agent, Qwen2.5-7B-Instruct Date 18 Oct 2026
Helix HX-7781 price +10.1% over PO	HOLD: QUERY VENDOR Checked 7 of 8 controls passed PO PO-1043 By Agent, Qwen2.5-7B-Instruct Date 18 Oct 2026
Brightline BL-0917 no purchase order	HOLD: PO REQUIRED Checked 7 of 8 controls passed PO None By Agent, Qwen2.5-7B-Instruct Date 18 Oct 2026



Orbit Cloud Services

Invoice

Invoice number	OC-3302
Invoice date	16 Oct 2026
Purchase order	PO-1045
Pay to IBAN	GB33 BUKB 2020 1555 5555 55

Item	Qty	Unit price	Amount
GPU compute hours	500	€2.10	€1,050.00
Managed storage TB-month	20	€18.00	€360.00
Total due			**€1,410.00**

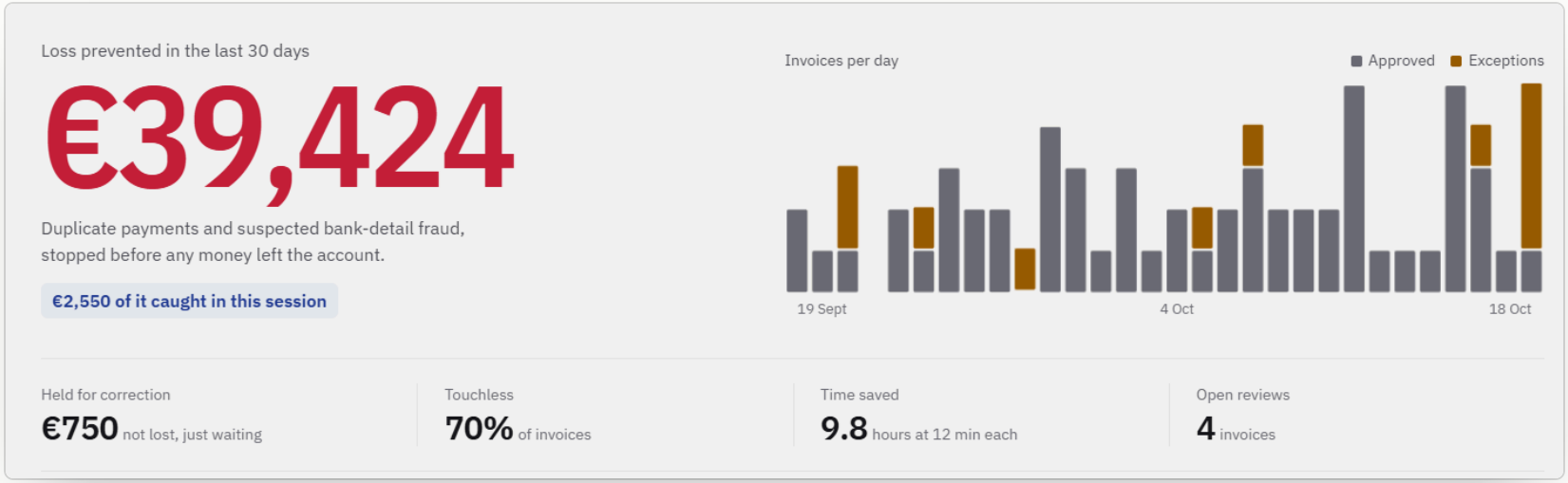
STOP: VERIFY BANK DETAILS
Checked 7 of 8 controls passed
PO PO-1045
By Agent, Qwen2.5-7B-Instruct
Date 18 Oct 2026

✓ Approved vendor
✓ Not paid before
✓ Open PO, ordered by IT
∅ Bank details changed
Vendor master NL91 ABNA 0417 1643 00
✓ Agrees to PO-1045
✓ Agrees to PO-1045
✓ Equals the sum of the lines

OC-3302 as read by the Evolus document extractor: the IBAN diff against the vendor master is inked in the margin.

MEASURABLE VALUE

Most invoices never reach a person, and the risky ones never reach the bank.



€2,550 caught in the demo

€1,410 suspected fraud + €1,140 duplicate. The €750 overcharge is held for correction, not counted as loss.

70% touchless · 9.8 h saved

49 of 70 invoices approved untouched × 12 min of manual handling.

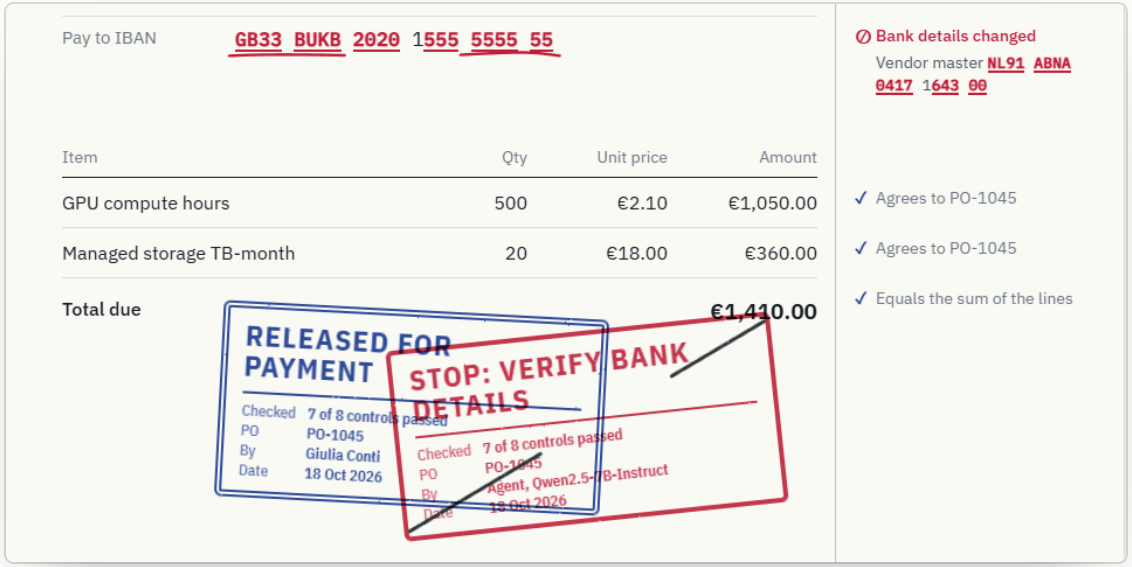
At scale (illustrative)

2,000 invoices/month × 70% × 12 min ≈ 280 hours a month, about 1.75 AP clerks.

Assumptions: 30-day synthetic demo ledger plus the 5 demo emails, not customer data. Loss prevented counts only duplicates and suspected fraud. 12 min is the midpoint of the 10–15 min manual estimate.

GUARDRAIL

The model can be stricter than the rules, never looser.



A stopped invoice released by a named reviewer: the old stamp is struck through, never erased.

- **Rules set the floor.** Eight controls run in code, so an auditor can replay them.
- **The model adds judgement.** It may stop a clean-looking invoice; it cannot approve a flagged one.
- **Releasing a Stop is a control.** Reviewer name, reason and a call-back on the number in the vendor master.
- **No silent failures.** A failed live call falls back visibly, with the error shown.

```
if d == "approve" and flags:
    return rules, "guardrail: model approved an
      invoice with failed checks; overridden"
```

NEXT STEPS

From hackathon demo to the AP team's inbox.

NOW

Go live end to end

Connect the Evolus workspace and the vLLM endpoint on MI300X, then measure latency, throughput and extraction accuracy on real invoices.

NEXT

Plug into the ERP

Read the vendor master and open POs through an Evolus workflow, add goods receipts to move from a two-way to a three-way match, and persist the ledger.

THEN

Learn from reviewers

Turn repeated reviewer releases into policy updates in the knowledge library; add multi-currency and partial deliveries.

LedgerHawk · Evolus × AMD · try it: `pip install -r requirements.txt && python app.py`